

CREDIXA

A Rapid Loan Disbursement Platform & Next-Gen Education BNPL with Autonomous Underwriting

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1. Executive Summary & Problem Context

The Problem: Rising educational costs create severe barriers for deserving students. Traditional financial options require extensive physical paperwork and rely exclusively on historical credit bureau scores (e.g., CIBIL, FICO). This creates the **'thin-file dilemma'**: students with strong academic merit but zero borrowing history are systematically denied loans or subjected to 14–21 day approval delays.

The Credixa Solution: Credixa is a production-deployed Buy Now Pay Later (BNPL) platform that democratizes credit access. It replaces archaic credit checks with an AI-driven, real-time behavioral underwriting engine that evaluates non-traditional data—banking cash flows, Debt-to-Income (DTI) ratios, income stability, and academic merit—to compute an inclusive 0–900 credit score called the **Omniscore** in under 60 seconds.

2. Unique Value Proposition (UVP)

Core UVP: “Credixa democratizes education financing by turning credit-invisible students into creditworthy borrowers through real-time (<60s) AI behavioral underwriting (0–900 Omniscore), Explainable AI (XAI) transparency, and zero-knowledge data privacy—replacing weeks of bureaucratic delays with instant, fair, and automated credit access.”

• **Key Differentiators:** Non-traditional behavioral scoring (zero bureau history required) • Instant sub-minute turnaround • Transparent pros/cons explainability • Bank-grade AES-256-GCM / HMAC encryption • Co-lending TSP regulatory alignment.

3. System Architecture & The 5 Pillars

Pillar	Technology Stack	Core Architectural Responsibilities
1. Frontend UI	React 19, Vite, Tailwind	Multi-tenant SPA (Student/Admin/Superadmin), 5-step applicant wizard, dynamic OCR correction modal, and real-time SSE stream listeners.
2. API Gateway	Node.js, Express, JWT, Redis	Central router managing dual-token auth (15m JWT + 7d HttpOnly cookie), RBAC authorization, token blacklist, and 01:00 AM daily cron scheduler.
3. AI Risk Engine	Python 3.11, FastAPI, XGBoost	Multi-page PDF extraction (pdfplumber + Tesseract OCR), 9-feature scaling, XGBoost ML scoring, and LLM Explainable AI dynamic pros/cons.
4. Database Vault	PostgreSQL 16 (Supabase)	11 relational 3NF tables, UUID keys, custom ENUMs, AES-256-GCM encrypted PII, deterministic HMAC indexes, and atomic ACID ledgers.
5. Session Cache	Redis 7 Alpine	Sliding-window IP rate limiting (5 req/15m), revoked token storage (bl_\$(ti)), and SHA-256 PDF text caching for fast inference.

4. AI Underwriting & The Omniscore Pipeline

- **9-Feature Underwriting Vector:** [avg_balance, overdrafts, gambling_flags, academic_score, dti_ratio, savings_rate, income_stability, fraud_flag, name_mismatch_flag].
- **ML Model & Score Translation:** Standardized via StandardScaler and classified by an **XGBoost Classifier** (n_estimators=300, max_depth=5, lr=0.05). Approval probability P(Approved) translates to a 0–900 Omniscore: Omniscore = round(P(Approved) × 900).
- **Explainable AI (XAI) & Drift Monitoring:** Generates dynamic, human-readable reasoning and Pros/Cons via LLMs/SHAP. evaluate_drift.py tracks **Population Stability Index (PSI)** to flag data distribution shifts.
- **Dual-Layer OCR Fallback:** Native text extraction via pdfplumber (<200ms) with seamless fallback to pytesseract image OCR and local regex heuristics (99.9% uptime).

5. Multi-Layer Security, Cryptography & Compliance

Security Domain	Technical Implementation	Security & Regulatory Benefit
PII Encryption	AES-256-GCM (iv:authTag:cipher)	Guarantees zero-knowledge protection for sensitive Aadhaar & PAN numbers at rest.
O(1) Deduplication	HMAC-SHA256 Deterministic Hash	Allows instant duplicate account detection via indexed column without full table decryption.
PII Redaction	Regex Scrubber (pii_redactor.py)	Strips government IDs, phones, and bank accounts before sending prompts to external LLMs (DPDP / GDPR compliant).
Dual-Token JWT	15m JWT + 7d HttpOnly Cookie	Prevents XSS token theft; enables instant server-side revocation via Redis token blacklist.
Inter-Service ML Auth	x-api-key Header Validation	Strictly authenticates Gateway-to-Risk-Engine REST calls, isolating ML assets.

6. Quantitative Performance Metrics & Validation

94.2%	< 2.8s	< 60s	99.9%	100%
XGBoost Model Accuracy	AI Inference Latency	End-to-End Decision Time	Pipeline System Uptime	Tamper Detection Rate

End-to-End Test Case Validation Matrix:

Scenario / Applicant	Input Documents & Risk Triggers	AI Underwriting Evaluation & Decision	Result
TC-01: Rahul Sharma (Happy Path)	Clean 6-month statements, 95.2% GPA, Rs. 65,000 balance, DTI 15.2%.	Omniscore: 850 (LOW_RISK) Instant Rs. 2,50,000 loan approval & transparent zero-interest schedule.	PASSED
TC-02: Kamran Khan (High DTI Risk)	Rs. 45k monthly sales, Rs. 41k existing loan EMIs (DTI: 91.1%), Rs. 500 liquidity.	Omniscore: 450 (HIGH_RISK) Automated rejection with clear debt burden explanation.	PASSED
TC-03: Harpreet Singh (Tamper & Fraud)	Manipulated PDF with Rs. 10L jump without credit entry; forged 30-Feb date.	Omniscore: 12.0 (FRAUD_LOCK) Immediate application lock; Critical fraud alert dispatched to admin.	PASSED
TC-04: Rhea D'Souza (OCR Discrepancy)	Extremely blurry fee structure JPEG with OCR confidence < 40%.	Flagged for Re-upload Pristine PDF uploaded -> parsed Rs. 3,80,000 -> Approved.	PASSED

7. Regulatory Model, Financial Automation & Future Roadmap

- **Regulatory TSP Model:** Credixa operates as a **Lending Service Provider (LSP) / Technology Service Provider (TSP)** in partnership with licensed Tier-2 NBFCs and banks. All capital disbursements flow directly from partner bank escrow accounts, while Credixa generates revenue from platform fees and origination fees (1.5–3%)—ensuring 100% compliance with digital lending regulations.
- **Financial Automation:** Repayments use reducing-balance amortization. A daily background cron job (0 1 * * *) inspects overdue schedules, applies flat Rs. 500 late fees, and auto-escalates to DEFAULTED status after 3 consecutive missed payments.
- **Conclusion & Future Work:** Credixa validates that AI behavioral underwriting can safely replace legacy credit bureaus, slashing loan turnaround from 3 weeks to under 60 seconds. Future roadmap includes: (1) IndiaStack Account Aggregator (Sahamati/Plaid) API integration; (2) React Native mobile application; (3) Multi-region AWS Fargate deployment.